

TCS IPA PREPARATION 2025

UI (HTML • CSS • JavaScript)

Rapid Revision Handbook

Everything you need the day before the exam —

tags, selectors, properties, methods, syntax, comparisons, MCQs & memory tricks

- ✓ Every HTML tag from the TCS IPA UI syllabus, explained concisely
- ✓ CSS properties, selectors and layout — exam-focused
- ✓ JavaScript methods, DOM, events and browser objects
- ✓ 30+ comparison tables for the most confused pairs
- ✓ 150+ practice MCQs in TCS IPA style, with explained answers
- ✓ A 5-page rapid-fire revision section for the final hours

Compiled from the official TCS IPA UI/UX study material, corrected and reorganised for revision.

How to Use This Book

This handbook replaces the original TCS IPA UI/UX PDF for revision. Each chapter gives you the concept in a few lines, a reference table, the exact examples you're expected to know, an exam tip, and a common mistake to avoid — no filler.

- First pass (2–3 days before exam): Read all chapters fully, experiment with small code snippets in a browser.
- Second pass (day before exam): Read only the shaded EXAM TIP / MISTAKE boxes and the Comparison Tables chapter.
- Final hours: Read only the Rapid Revision section at the end — it condenses the entire book to 5 pages.
- Practice: Attempt the 150+ MCQs in the Practice section; check answers only after attempting.

Table of Contents

HTML Cheat Sheet	Chapter 15 — Variables, Operators & Functions
Chapter 1 — HTML Basics	Chapter 16 — Arrays & Objects
Chapter 2 — HTML Formatting	Chapter 17 — Loops & Conditions
Chapter 3 — Lists	Chapter 18 — DOM & Events
Chapter 4 — Tables	Chapter 19 — String, Math & Date Methods
Chapter 5 — Forms & Input	Chapter 20 — Browser Objects & Error Handling
Chapter 6 — Media (Images, Audio, Video)	TCS IPA Important Points
Chapter 7 — Semantic HTML & HTML5	Complete HTML Summary Table
Chapter 8 — CSS Introduction & Selectors	Complete CSS Summary Table
Chapter 9 — Colors, Backgrounds & Borders	Complete JavaScript Summary Table
Chapter 10 — Box Model (Margin, Padding, Width)	Comparison Tables
Chapter 11 — Display, Position & Flexbox	Practice MCQs
Chapter 12 — Fonts, Text & Links	Rapid Revision — Final Hours
Chapter 13 — CSS Lists, Tables & Pseudo	
Chapter 14 — JavaScript Basics	

HTML Cheat Sheet

One-page quick reference

Category	Tag / Attribute	Purpose
Structure	<!DOCTYPE html>	Declares HTML5 document
Structure	<html> <head> <body>	Root, metadata, content
Headings	<h1>...<h6>	h1=largest, h6=smallest
Text	<p> <hr>	Paragraph, line break, horizontal rule
Formatting	 	Bold / semantically important
Formatting	<i> 	Italic / emphasised
Formatting	<u> <ins>	Underline / inserted text
Formatting	<s> 	Strikethrough / deleted
Formatting	<mark> <sup> <sub>	Highlight, superscript, subscript
Formatting	<pre> <code>	Preformatted / inline code
Formatting	<blockquote> <q>	Long / inline quotation
Lists		Unordered, ordered, list item
Lists	<dl> <dt> <dd>	Definition list, term, description
Table	<table> <tr> <th> <td>	Table, row, header cell, data cell
Table	<caption> <thead> <tbody> <tfoot>	Table caption, head/body/foot groups
Form	<form> <input> <label>	Form container, input, label
Form	<textarea> <select> <option>	Multi-line input, dropdown, option
Form	<button> <fieldset> <legend>	Button, field group, group label
Media		Image
Media	<audio controls> <video controls>	Audio / video player
Media	<source> <iframe>	Media source, inline frame
Links	<a href target>	Hyperlink
Semantic	<header> <nav> <main>	Page header, navigation, main content
Semantic	<section> <article> <aside>	Section, standalone article, sidebar
Semantic	<footer> <figure> <figcaption>	Footer, figure, caption
HTML5	<canvas> <svg>	Bitmap drawing, vector graphics
HTML5	<details> <summary> <dialog>	Expandable widget, dialog

Category	Tag / Attribute	Purpose
HTML5	<progress> <meter> <datalist>	Progress bar, gauge, input suggestions
Meta	<meta charset name content>	Document metadata
Global Attrs	id class style title	Identity, class, inline style, tooltip
Input types	text password email number	Standard input types
Input types	date checkbox radio file	Date picker, toggle, file upload
Input types	range color search tel url	Slider, colour, search, phone, URL

Chapter 1 — HTML Basics

Document structure, headings, paragraphs, links, images

HTML (HyperText Markup Language) is the standard language for creating web pages. Every HTML document is a tree of nested elements — tags enclosed in angle brackets. Most tags have an opening and a closing form; a few (void elements) are self-closing.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Page Title</title>
  </head>
  <body>
    <h1>Hello World</h1>
  </body>
</html>
```

Tag	Purpose	Exam Importance
<!DOCTYPE html>	Declares HTML5 to the browser	High
<html>	Root element of the page	High
<head>	Contains metadata — not displayed	High
<body>	All visible page content	High
<title>	Browser tab title (inside <head>)	Medium
<meta charset="UTF-8">	Character encoding declaration	Medium
<h1>–<h6>	Headings: h1 largest, h6 smallest	High
<p>	Paragraph of text	High
 	Line break (void, no closing tag)	High
<hr>	Horizontal rule / thematic break	Medium
	Hyperlink — href is mandatory	High
target="_blank"	Opens link in a new tab	High
	Image — void element, alt is mandatory	High

★ EXAM TIP

target="_blank" opens in a new tab. target="_self" (default) opens in same tab. This is a favourite MCQ. Also: img uses src not href.

■ COMMON MISTAKE

Forgetting `alt=""` on images — it is mandatory for accessibility and many MCQs test it. Also: `<br>` has NO closing tag — writing `</br>` is wrong.

Chapter 2 — HTML Formatting

Text-level semantic and presentational tags

Formatting tags fall into two groups: semantic (carry meaning) and presentational (only affect appearance). TCS IPA heavily tests the semantic vs visual distinction.

Tag	Renders as	Meaning / Purpose	Exam Importance
	Bold	Presentational bold, no extra meaning	High
	Bold	Semantically important text	High
<i>	Italic	Presentational italic	High
	Italic	Emphasised / stressed text	High
<u>	Underline	Underline (presentational)	Medium
<ins>	Underline	Inserted text (semantic)	Medium
<s>	Strikethrough	Presentational strike	Medium
	Strikethrough	Deleted text (semantic)	Medium
<mark>	Yellow highlight	Highlighted / relevant text	Medium
<sup>	Superscript	e.g. x ² → x²	Medium
<sub>	Subscript	e.g. H ₂ O → H₂O	Medium
<pre>	Monospace	Preserves whitespace & newlines	High
<code>	Monospace	Inline code snippet	Medium
<blockquote>	Indented block	Long quotation	High
<q>	Inline with quotes	Short inline quotation	Medium
<marquee>	Scrolling text	Scrolling effect (deprecated but TCS asks)	High
<abbr>	Dotted underline	Abbreviation with tooltip	Low

```
<strong>Important</strong> vs <b>Bold</b>
<em>Emphasis</em> vs <i>Italic</i>
<ins>Inserted</ins> – appears underlined
<del>Deleted</del> – appears strikethrough
<marquee>This scrolls!</marquee>
```

★ EXAM TIP

TCS IPA asks: Which tag emphasises text? → . Which tag makes text semantically important? → . Which tag makes text scroll? → <marquee>.

■ COMMON MISTAKE

`<b>` and `<strong>` both bold text visually but `<strong>` carries semantic weight. Similarly `<i>` vs `<em>`. The TCS IPA PDF tests both forms — learn both.

Chapter 3 — Lists

Unordered, ordered, definition lists

Tag	Purpose	Key Attributes	Exam Importance
	Unordered (bulleted) list	type: disc/circle/square	High
	Ordered (numbered) list	type: 1/A/a/I/i, start, reversed	High
	List item (inside ul or ol)	value (in ol)	High
<dl>	Definition list	—	Medium
<dt>	Definition term	—	Medium
<dd>	Definition description	—	Medium

```
<!-- Unordered list -->
<ul type="square">
  <li>Item A</li>
  <li>Item B</li>
</ul>

<!-- Ordered list with Roman numerals -->
<ol type="I">
  <li>First</li>
  <li>Second</li>
</ol>
```

★ EXAM TIP

Ordered list types: type='1' (default numbers), type='A' (uppercase letters), type='a' (lowercase), type='I' (uppercase Roman), type='i' (lowercase Roman). TCS IPA directly asks about Roman numeral syntax.

■ COMMON MISTAKE

Unordered list bullet types: disc (default), circle, square. Triangle is NOT a valid type — a very common wrong-answer option.

Chapter 4 — Tables

Rows, columns, spanning,
captions

HTML tables are built with nested elements. Every table cell is either `<th>` (header) or `<td>` (data). The `<caption>` tag belongs inside `<table>`.

Tag	Purpose	Key Attributes	Exam Importance
<code><table></code>	Table container	border, width	High
<code><tr></code>	Table row	—	High
<code><th></code>	Header cell (bold, centred by default)	colspan, rowspan	High
<code><td></code>	Data cell	colspan, rowspan	High
<code><caption></code>	Table caption (above or below table)	—	High
<code><thead></code>	Groups header rows	—	Medium
<code><tbody></code>	Groups body rows	—	Medium
<code><tfoot></code>	Groups footer rows	—	Medium

```
<table border="1">
  <caption>Student Marks</caption>
  <tr><th>Name</th><th>Marks</th></tr>
  <tr><td>Alice</td><td>95</td></tr>
  <tr><td colspan='2'>Total: 95</td></tr>
</table>
```

★ EXAM TIP

`<caption>` is used with tables — not images. 'Which tag is used to add a title to a table?' → `<caption>`. This is repeatedly asked.

■ COMMON MISTAKE

`colspan` merges columns horizontally. `rowspan` merges rows vertically. They are attributes of `<td>` or `<th>`, not of `<table>`.

Chapter 5 — Forms & Input

Form elements, input types, GET vs POST

Forms collect user data and send it to a server. The `<form>` action attribute specifies the destination URL; method specifies how data is sent (GET vs POST).

Element	Purpose	Key Attributes	Exam Importance
<code><form action method></code>	Form container	action, method (GET/POST)	High
<code><input type="text"></code>	Single-line text input	name, value, placeholder, required	High
<code><input type="password"></code>	Password (masked input)	name, required	High
<code><input type="email"></code>	Email — validates format	required	High
<code><input type="number"></code>	Numeric input	min, max, step	High
<code><input type="checkbox"></code>	Checkbox toggle	checked	High
<code><input type="radio"></code>	Radio button (one-of-group)	name (must match group)	High
<code><input type="submit"></code>	Form submit button	value (button label)	High
<code><input type="file"></code>	File upload picker	accept	Medium
<code><input type="date"></code>	Date picker	min, max	High
<code><input type="range"></code>	Slider	min, max, step	Medium
<code><input type="color"></code>	Colour picker	—	Medium
<code><input type="hidden"></code>	Hidden field (not shown)	name, value	Medium
<code><textarea></code>	Multi-line text input	rows, cols	High
<code><select> + <option></code>	Dropdown (combo box)	selected	High
<code><button></code>	Clickable button	type (submit/reset/button)	High
<code><label></code>	Label linked to input	for (matches input id)	High
<code><fieldset> + <legend></code>	Groups related inputs	—	Medium
<code><datalist></code>	Autocomplete suggestions	id linked to input list attr	Medium

★ EXAM TIP

Dropdown (combo box) = `<select>`. Options inside = `<option>`. 'Which tag creates a dropdown?' → `<select>`. TCS IPA asks this repeatedly.

■ COMMON MISTAKE

datetime is NOT a valid HTML5 input type — datetime-local is. TCS IPA asks: 'Which is an invalid HTML5 field?' → Datetime. Don't mix up date, datetime-local, month, week.

Tag	Purpose	Key Attributes	Exam Importance
	Embed image (void tag)	src, alt, width, height	High
<audio>	Embed audio player	src, controls, autoplay, loop, muted	High
<video>	Embed video player	src, controls, width, height, poster	High
<source>	Alternative media source	src, type	Medium
<iframe>	Embed another HTML page	src, width, height, title	Medium
<figure>	Self-contained media content	—	Medium
<figcaption>	Caption for <figure>	—	Medium
<canvas>	Scriptable bitmap drawing area	id, width, height	High
<svg>	Inline vector graphics	viewBox, width, height	High

```

```

```
<audio controls>
  <source src="song.mp3" type="audio/mpeg">
</audio>
```

```
<video controls width="400">
  <source src="clip.mp4" type="video/mp4">
</video>
```

★ EXAM TIP

<svg> draws vector graphics (paths, shapes). <canvas> draws raster graphics via JavaScript scripting. TCS IPA asks which tag 'draws graphics on the fly via scripting' → <canvas>.

■ COMMON MISTAKE

opacity is NOT a valid HTML attribute for . Setting opacity in HTML inline style works (style='opacity:0.25') but not as a direct attribute. The image still displays — just semi-transparent.

Chapter 7 — Semantic HTML & HTML5

Layout tags, new HTML5 elements

Semantic tags describe the meaning of content, not just its appearance. HTML5 introduced many semantic elements that replaced generic `<div>` blocks.

Tag	Purpose	Exam Importance
<code><header></code>	Page or section header (logo, nav)	High
<code><nav></code>	Primary navigation links	High
<code><main></code>	Main content area — one per page	High
<code><section></code>	Thematic grouping of content	High
<code><article></code>	Self-contained piece (blog post, news item)	High
<code><aside></code>	Sidebar or tangentially related content	High
<code><footer></code>	Page or section footer	High
<code><figure></code>	Self-contained media with optional caption	Medium
<code><figcaption></code>	Caption for <code><figure></code>	Medium
<code><details></code>	Expandable/collapsible widget	Medium
<code><summary></code>	Visible heading of <code><details></code>	Medium
<code><dialog></code>	Modal or popup dialog	Medium
<code><progress></code>	Progress bar (0 to max)	Medium
<code><meter></code>	Scalar gauge within a range	Low
<code><time></code>	Machine-readable date/time	Low
<code><mark></code>	Highlighted / relevant text	Medium
<code><canvas></code>	Scriptable bitmap drawing area	High
<code><svg></code>	Inline vector graphics container	High
<code><datalist></code>	Autocomplete suggestions for <code><input></code>	Medium
<code><output></code>	Result of a calculation	Low

★ EXAM TIP

All of these were added in HTML5: `<main>`, `<header>`, `<footer>`, `<article>`, `<section>`, `<aside>`, `<nav>`, `<figure>`, `<figcaption>`, `<details>`, `<summary>`, `<dialog>`, `<progress>`, `<canvas>`, `<svg>`. TCS IPA asks: 'Which elements were added in HTML5?' — the answer is ALL of the above.

■ COMMON MISTAKE

`<article>` is a self-contained, independently distributable piece of content. `<section>` is a thematic grouping that needs context from the surrounding page. Don't swap them.

CSS (Cascading Style Sheets) controls the visual presentation of HTML. There are three ways to include CSS: inline (style attribute), internal (<style> in <head>), and external (linked .css file in <head>).

Selector	Syntax	Selects	Example	Exam Importance
Element	p { }	All <p> elements	p { color:red; }	High
Class	.name { }	Elements with class	.box { }	High
ID	#name { }	Element with id	#header { }	High
Universal	* { }	Every element	* { margin:0; }	Medium
Descendant	div p { }	<p> inside any <div>	div p { }	High
Child	div > p { }	<p> direct child of <div>	div > p { }	Medium
Adjacent	h1 + p { }	<p> immediately after <h1>	h1 + p { }	Medium
Attribute	[type=text]	Input with type=text	[type='text'] { }	Medium
Grouped	h1, h2 { }	Both h1 and h2	h1, h2 { }	High
Element.class	h1.intro { }	h1 with class intro	h1.xplore { }	High

```

/* External – link in <head> */
<link rel="stylesheet" href="style.css">

/* Internal – inside <head> */
<style> p { color: blue; } </style>

/* Inline – on the element */
<p style="color: red;">Red text</p>

```

★ EXAM TIP

CSS is placed in the <head> section when using external or internal styles. 'Where is the correct place to refer to an external stylesheet?' → In the <head> section. Priority: inline > internal > external.

■ COMMON MISTAKE

Class selector uses . (dot). ID selector uses # (hash). Writing xplore.h1 is wrong — the correct element.class selector is h1.xplore.

Chapter 9 — Colors, Backgrounds & Borders

CSS colour formats, background properties, border shorthand

Property	Purpose	Example	Exam Importance
color	Text colour	color: red; color: #FF0000; color: rgb(255,0,0)	High
background-color	Element background colour	background-color: yellow	High
background-image	Background image	background-image: url('bg.jpg')	High
background-repeat	Repeat behaviour	background-repeat: no-repeat	Medium
background-size	Size of background image	background-size: cover	Medium
background-attachment	Scroll or fixed	background-attachment: fixed	Medium
border	Shorthand border	border: 2px solid DodgerBlue	High
border-color	Border colour only	border-color: red	Medium
border-width	Border width only	border-width: 3px	Medium
border-style	Border style	border-style: solid / dashed / dotted	High
border-radius	Rounded corners	border-radius: 10px	Medium
outline	Border outside border-box	outline: 2px solid red	Medium

```
h1 { border: 2px solid DodgerBlue; } /* shorthand: width style color */
body { background-color: yellow; }
div { background-image: url('bg.jpg'); background-repeat: no-repeat; }
```

★ EXAM TIP

Correct border shorthand: border: 2px solid DodgerBlue — width, then style, then colour in that order. 'border-line' is NOT a valid property. border-render is NOT a valid property.

■ COMMON MISTAKE

White, Gray, and Black all have equal amounts of R, G, B (just different values). #oooff uses letter 'o' instead of zero '0' — invalid hex. Hex colours use 0–9 and A–F only.

Chapter 10 — Box Model — Margin, Padding, Width & Height

The CSS box model explained

Every HTML element is a rectangular box. From outside in: margin → border → padding → content. Understanding this is critical for layout questions.

Property	Purpose	Example	Exam Importance
margin	Space OUTSIDE the border	margin: 10px; margin: 10px 20px 10px 20px	High
margin: auto	Centre block element horizontally	margin: 0 auto	High
padding	Space INSIDE the border, around content	padding: 15px	High
width	Content area width	width: 300px; width: 50%	High
height	Content area height	height: 200px	High
max-width	Maximum width constraint	max-width: 800px	Medium
min-width	Minimum width	min-width: 200px	Medium
box-sizing	Include border+padding in width	box-sizing: border-box	Medium

```
/* Shorthand: top right bottom left (clockwise) */  
margin: 10px 20px 30px 40px;
```

```
/* float is NOT a property applicable to paragraphs in TCS MCQs */  
/* Use margin, border, font-family on <p> */
```

★ EXAM TIP

Margin = space OUTSIDE the element (between elements). Padding = space INSIDE the element (between content and border). display:none completely removes from layout; visibility:hidden hides but keeps the space.

■ COMMON MISTAKE

float is NOT applicable to a paragraph per TCS IPA MCQs. The MCQ answer is: 'Which property is not applicable for a paragraph?' → float.

Chapter 11 — Display, Position & Flexbox

Layout control

Property / Value	Behaviour	Exam Importance
display: block	Takes full width, starts on new line	High
display: inline	Flows with text, no width/height	High
display: inline-block	Inline flow but accepts width/height	High
display: none	Removes element from layout entirely	High
display: flex	Enables flexbox on container	High
visibility: hidden	Hides element but keeps layout space	High
position: static	Default — normal document flow	Medium
position: relative	Offset from its normal position	High
position: absolute	Positioned relative to nearest positioned ancestor	High
position: fixed	Fixed to viewport, doesn't scroll	High
position: sticky	Sticks after scrolling to threshold	Medium
z-index	Stack order (higher = in front)	Medium
float: left / right	Float element beside text	Medium
clear	Stop floating around element	Medium
flex-direction	Row or column flex direction	Medium
justify-content	Align on main axis	Medium
align-items	Align on cross axis	Medium

★ EXAM TIP

display: block causes a line break between elements. 'What does display:block result in?' → A line break between elements. <div> is block-level; is inline — a very common comparison question.

■ COMMON MISTAKE

display:none vs visibility:hidden — both hide the element, but display:none removes the space it occupied while visibility:hidden leaves an invisible gap. This exact comparison is tested.

Property	Purpose	Example	Exam Importance
font-family	Typeface	font-family: Arial, sans-serif	High
font-size	Size (px, em, rem, %)	font-size: 16px	High
font-weight	Boldness	font-weight: bold / 700	High
font-style	Italic or normal	font-style: italic	High
text-align	Horizontal alignment	text-align: center	High
text-decoration	Underline, overline, etc.	text-decoration: underline	High
text-transform	Capitalisation	text-transform: uppercase	Medium
line-height	Space between lines	line-height: 1.5	Medium
letter-spacing	Space between chars	letter-spacing: 2px	Medium
color	Text colour	color: #333	High
a:link	Unvisited link	a:link { color: blue; }	High
a:visited	Visited link	a:visited { color: purple; }	High
a:hover	Mouse over link	a:hover { color: red; }	High
a:active	Link being clicked	a:active { color: green; }	Medium

```
/* Font size – correct syntax */
h2 { font-size: 200px; } /* pixels */
h2 { font-size: 200%; } /* percentage – also valid */
/* NOT valid: font-size: 200pixels; or font-size: 200; */
```

★ EXAM TIP

Link pseudo-class order matters: :link → :visited → :hover → :active (LoVeHAtE mnemonic). font-size: 200px and 200% are both valid; '200pixels' and bare '200' are not.

■ COMMON MISTAKE

em is relative to the parent's font size. rem is relative to the root (<html>) font size. px is an absolute pixel unit. % is relative to the containing element.

Chapter 13 — CSS Lists, Tables, Pseudo Classes & Comments

Remaining CSS essentials

Property / Selector	Purpose	Example	Exam Importance
list-style-type	Bullet/number type	list-style-type: square	High
list-style-image	Custom bullet image	list-style-image: url('b.png')	Low
border-collapse	Collapse table borders	border-collapse: collapse	High
border-spacing	Space between table cells	border-spacing: 5px	Medium
:hover	Style on mouse over	a:hover { color:red; }	High
:focus	Style when element has focus	input:focus { outline:blue; }	High
:first-child	First child of parent	p:first-child { }	Medium
:nth-child(n)	Every nth child	tr:nth-child(2n) { }	Medium
:checked	Checked checkbox/radio	input:checked { }	Medium
::before	Insert content before element	p::before { content:'>'; }	Medium
::after	Insert content after element	p::after { content:'.'; }	Medium
::first-letter	Style first letter	p::first-letter { font-size:2em; }	Medium
/* comment */	CSS comment syntax	/* this is a comment */	High

★ EXAM TIP

CSS comments use `/* */`. The block `p { /* comment */ color: red; }` is valid. `p { // comment color: red; }` and `p { #comment }` are INVALID. A favourite MCQ.

■ COMMON MISTAKE

Pseudo-classes (`:hover`) target element STATES. Pseudo-elements (`::before`, `::after`) target specific PARTS of the element. Single colon (`:`) is pseudo-class; double colon (`::`) is pseudo-element.

Chapter 14 — JavaScript Basics

Adding scripts, data types, type coercion

JavaScript is the scripting language of the web. It runs in the browser (client-side) via a script engine. TCS IPA tests: where to place `<script>`, data types, NaN, undefined, and type coercion.

Concept	Detail	Exam Importance
<code><script></code> tag	Use <code><script type='text/javascript'></script></code>	High
Placement	Preferably before <code></body></code> or in <code><head></code>	High
Data types	number, string, boolean, undefined, null, object, symbol	High
NaN	Not a Number — result of invalid arithmetic (e.g. <code>NaN * 3 = NaN</code>)	High
undefined	Declared variable with no value assigned	High
null	Intentional absence of value (<code>typeof null = 'object'</code> — legacy quirk)	High
typeof	Returns type as string: <code>typeof 5 → 'number'</code>	High
true/false	Boolean values — NOT numeric 0/1 in JS	High
Division by zero	Returns Infinity, not an error — NOT treated as an error	High
Missing semicolons	Syntax leniency — ASI inserts them, not an error per TCS	High
LiveWire / server-side JS	Server-side JavaScript platform by Netscape	Medium

```
<script type='text/javascript'>
  var x;           // undefined
  x = myFunction(NaN, 3); // NaN * 3 = NaN
  document.getElementById('result').innerHTML = x; // NaN
  function myFunction(a, b) { return a * b; }
</script>
```

★ EXAM TIP

Division by zero in JavaScript is NOT an error — it returns Infinity. Missing semicolons are NOT errors (ASI). These are the two 'not an error' options in TCS IPA MCQs.

■ COMMON MISTAKE

true and false in JS are NOT numeric values — they are boolean. This is directly tested: 'Which statement is correct? → true and false are nonnumeric values.'

Chapter 15 — Variables, Operators & Functions

var/let/const, operators, function syntax

Keyword	Scope	Re-declare	Re-assign	Hoisted	Exam Importance
var	Function/global	Yes	Yes	Yes (as undefined)	High
let	Block	No	Yes	No (TDZ)	High
const	Block	No	No	No (TDZ)	High

Operator	Meaning	Example	Result	Exam Importance
==	Loose equality (type coercion)	5 == '5'	true	High
===	Strict equality (value AND type)	5 === '5'	false	High
!=	Loose not-equal	5 != '5'	false	High
!==	Strict not-equal	5 !== '5'	true	High
&&	Logical AND	true && false	false	High
	Logical OR	true false	true	High
!	Logical NOT	!true	false	High
typeof	Type of value	typeof 'hello'	'string'	High
^	Bitwise XOR (NOT power)	100 + 5 ^ 3	225 (XOR, not 5^3)	High

```
/* Function declaration */
function greet(name) { return 'Hello, ' + name; }

/* Arrow function */
const greet = (name) => 'Hello, ' + name;

/* eval() – evaluates a string as JS code */
var res = eval('12 + 8'); // res = 20
```

★ EXAM TIP

=== checks value AND data type. 'The statement x===y implies both are equal in value and data type' → Correct. x == y only checks value (with coercion). This is the most repeated JS MCQ.

■ COMMON MISTAKE

^ in JS is bitwise XOR, NOT the power operator. $100 + 5 \wedge 3 = (100 + 5) \wedge 3 = 105 \wedge 3 = 106$... actually let's be precise: JS precedence: + before ^, so $105 \wedge 3 = 106$. TCS IPA answer: 225 (based on their working: $100+5\wedge 3 = 100+125 = 225$ — treating ^ as power in their context). Follow their PDF answer.

Chapter 16 — Arrays & Objects

Array creation, methods, object syntax

Method / Property	Purpose	Example	Exam Importance
<code>[]</code> literal	Array creation	<code>var a = [1,2,3]</code>	High
<code>new Array()</code>	Array with 'new' keyword	<code>var a = new Array(1,2,3)</code>	High
<code>length</code>	Number of elements	<code>a.length → 3</code>	High
<code>push()</code>	Add to end	<code>a.push(4)</code>	High
<code>pop()</code>	Remove from end	<code>a.pop()</code>	High
<code>shift()</code>	Remove from start	<code>a.shift()</code>	Medium
<code>unshift()</code>	Add to start	<code>a.unshift(0)</code>	Medium
<code>indexOf()</code>	Find index of element	<code>a.indexOf(2) → 1</code>	High
<code>splice()</code>	Add/remove at index	<code>a.splice(1,1,'x')</code>	Medium
<code>slice()</code>	Extract sub-array	<code>a.slice(0,2)</code>	Medium
<code>forEach()</code>	Iterate each element	<code>a.forEach(fn)</code>	High
<code>map()</code>	Transform elements	<code>a.map(x => x*2)</code>	High
<code>filter()</code>	Filter elements	<code>a.filter(x => x>2)</code>	High
<code>reduce()</code>	Reduce to single value	<code>a.reduce((s,x) => s+x, 0)</code>	Medium
<code>sort()</code>	Sort in place	<code>a.sort()</code>	Medium
<code>join()</code>	Array to string	<code>a.join(',') → '1,2,3'</code>	Medium

```
var arr = [1, 2, 3];
var sum = 0;
arr.forEach(function(ele) { sum += ele; });
// sum = 6

var obj = { name: 'Alice', age: 25 };
console.log(obj.name); // Alice
```

★ EXAM TIP

Arrays in JS are defined with `[]` (list form). Can also use `'new Array(...)'`. Both are valid. 'I cannot create a JS array using the new keyword' → FALSE.

■ COMMON MISTAKE

`forEach` modifies the external sum variable via closure. The variable must be declared OUTSIDE the `forEach` callback to persist — a common output-tracing question.

Chapter 17 — Loops & Conditions

if/else, switch, for, while, do-while

Construct	Syntax	Use case	Exam Importance
if / else if / else	if (cond) {} else if (cond) {} else {}	Decision branching	High
switch / case	switch(x) { case v: break; default: }	Multiple fixed values	High
for	for (init; cond; update) {}	Known iterations	High
while	while (cond) {}	Unknown iterations, test first	High
do...while	do {} while (cond);	Test AFTER first iteration	High
for...in	for (key in obj) {}	Iterate object keys	Medium
for...of	for (val of arr) {}	Iterate iterable values	Medium
break	break;	Exit loop immediately	High
continue	continue;	Skip to next iteration	High

```
for (var i = 0; i < 3; i++) { console.log(i); } // 0 1 2

var x = 5;
switch(x) {
  case 5: console.log('Five'); break;
  default: console.log('Other');
}

if (a !== null) { /* a is NOT null */ }
```

★ EXAM TIP

To check if variable 'a' is NOT null: `if (a !== null)` — strict inequality. `if (a != null)` also works but checks loosely. The TCS IPA answer is `a !== null`.

■ COMMON MISTAKE

`do...while` runs the body at LEAST once — the test is at the end. `while` may never run if condition is false from the start.

Chapter 18 — DOM & Events

Document Object Model, selectors,
event handling

The DOM is a tree representation of the HTML document. JavaScript can read and modify any node in this tree to update the page dynamically.

Method / Property	Purpose	Example	Exam Importance
<code>getElementById(id)</code>	Select element by id	<code>document.getElementById('box')</code>	High
<code>getElementsByName(c)</code>	Select by class (HTMLCollection)	<code>document.getElementsByName('a')</code>	High
<code>querySelector(sel)</code>	First element matching CSS selector	<code>document.querySelector('.btn')</code>	High
<code>querySelectorAll(sel)</code>	All elements matching selector	<code>document.querySelectorAll('p')</code>	High
<code>.innerHTML</code>	Get/set HTML content inside element	<code>el.innerHTML = 'Hi'</code>	High
<code>.innerText</code>	Get/set text (visible text only)	<code>el.innerText = 'Hello'</code>	High
<code>.textContent</code>	Get/set all text including hidden	<code>el.textContent = 'Hello'</code>	Medium
<code>.value</code>	Get/set input field value	<code>input.value</code>	High
<code>.style.prop</code>	Set inline style	<code>el.style.color = 'red'</code>	High
<code>.classList</code>	Add/remove/toggle classes	<code>el.classList.add('active')</code>	Medium
<code>.setAttribute()</code>	Set any attribute	<code>el.setAttribute('href','url')</code>	Medium
<code>.getAttribute()</code>	Get attribute value	<code>el.getAttribute('id')</code>	Medium
<code>.firstChild.nodeValue</code>	Text of first child node	<code>para.firstChild.nodeValue='text'</code>	High
<code>addEventListener()</code>	Attach event handler	<code>el.addEventListener('click',fn)</code>	High
<code>onclick / onchange</code>	Inline event attribute	<code><button onclick='fn()></code>	High
<code>event.preventDefault()</code>	Stop default action	<code>e.preventDefault()</code>	Medium

★ EXAM TIP

To change text inside a paragraph DOM object `para1`: `para1.firstChild.nodeValue = 'New Text'`. Using `para1.value` or `para1.nodeValue` directly is WRONG.

■ COMMON MISTAKE

`getElementByClass()` does NOT exist — the correct method is `getElementsByName()` (plural Elements, and Class not Class+es). This typo is a common distractor in MCQs.

Chapter 19 — String, Math & Date Methods

Built-in JS object methods

Method	Purpose	Example	Output	Exam Importance
length	String length	'hello'.length	5	High
toUpperCase()	Convert to uppercase	'hi'.toUpperCase()	'HI'	High
toLowerCase()	Convert to lowercase	'HI'.toLowerCase()	'hi'	High
indexOf(x)	First index of x (-1 if not found)	'hello'.indexOf('l')	2	High
slice(s,e)	Extract substring	'hello'.slice(1,3)	'el'	High
substring(s,e)	Similar to slice	'hello'.substring(0,3)	'hel'	Medium
replace(a,b)	Replace first match	'hi'.replace('h','H')	'Hi'	High
split(sep)	String to array	'a,b'.split(',')	['a','b']	High
trim()	Remove leading/trailing spaces	' hi '.trim()	'hi'	Medium
includes(x)	Check if contains	'hello'.includes('ell')	true	Medium
Math.round(x)	Round to nearest integer	Math.round(4.6)	5	High
Math.floor(x)	Round down	Math.floor(4.9)	4	High
Math.ceil(x)	Round up	Math.ceil(4.1)	5	High
Math.max(a,b)	Largest value	Math.max(3,7)	7	High
Math.min(a,b)	Smallest value	Math.min(3,7)	3	High
Math.sqrt(x)	Square root	Math.sqrt(16)	4	Medium
Math.abs(x)	Absolute value	Math.abs(-5)	5	Medium
Math.random()	Random 0 to <1	Math.random()	0.xxx	Medium
new Date()	Current date/time	new Date()	Date obj	High
Date.getTime()	Milliseconds since Jan 1 1970	d.getTime()	number ms	High
Date.getFullYear()	Current year	d.getFullYear()	2025	High

★ EXAM TIP

JavaScript Date stores time as milliseconds since January 1, 1970 (Unix Epoch). 'How does JS store dates?'
→ number of milliseconds since January 1st, 1970.

■ COMMON MISTAKE

Invalid escape sequences in JS: `/e` is INVALID. Valid ones: `\n` (newline), `\r` (carriage return), `\t` (tab), `\b` (backspace), `\f` (form feed). The question format uses `/e`, `/r`, `/f`, `/b` — answer: `/e` is invalid.

Chapter 20 — Browser Objects & Error Handling

window, navigator, try/catch, error types

Object / Method	Purpose	Example	Exam Importance
window	Global browser object — top of hierarchy	window.alert('Hi')	High
window.alert()	Show popup alert	alert('Hello')	High
window.confirm()	OK/Cancel dialog	confirm('Sure?')	Medium
window.prompt()	Input dialog	prompt('Name?')	Medium
window.location	Current URL info	window.location.href	High
window.history	Browse history	history.back()	Medium
window.navigator	Browser/device info	navigator.userAgent	Medium
window.screen	Screen dimensions	screen.width	Low
document	DOM — represents the HTML document	document.title	High
try { } catch (e) { }	Handle errors gracefully	try { } catch(e) { }	High
finally { }	Always runs after try/catch	finally { cleanup(); }	Medium
throw	Throw custom error	throw new Error('Bad')	Medium
SyntaxError	Invalid JS syntax	—	High
ReferenceError	Undefined variable used	—	High
TypeError	Wrong type operation	—	High
RangeError	Value out of range	—	Medium

★ EXAM TIP

Error types in TCS IPA: Syntax errors, missing brackets = ERRORS. Division by zero = NOT an error (returns Infinity). Missing semicolons = NOT an error (ASI handles them). These are the most-asked distinctions.

■ COMMON MISTAKE

JavaScript runs using a SCRIPT ENGINE in the browser — this is TRUE. Server-side JavaScript was known as LiveWire (by Netscape/Navigator). Both facts are tested directly.

TCS IPA Important Points

Most asked, most forgotten, exam shortcuts

Most Asked HTML Tags

Tag	Why TCS Loves It
<marquee>	Only tag for scrolling text
 	Line break — void tag, NO closing tag
<blockquote>	How are quotations defined?
<a href target="_blank">	Opens in new tab vs new window
	src not href; alt is mandatory
<caption>	Table caption — always inside <table>
<select> + <option>	Dropdown / combo box
<svg> vs <canvas>	Vector vs scripted raster graphics
id attribute	Unique name — not class, not type
<ins>	Underline — appears underlined like <u>
<pre>	Special formatting tag — preserves whitespace
<ol type="I">	Ordered list with Roman numerals
<!DOCTYPE html>	Must be FIRST line of HTML document
<dd>	Definition data in a <dl> list
<script type='text/javascript'>	Correct way to insert JS in HTML

Most Asked CSS Properties

Property / Concept	Key Fact
border shorthand	Order: width style color (border: 2px solid DodgerBlue)
CSS comment	/* */ only — not // or #
Class selector	. (dot) not # (hash)
ID selector	# (hash) not . (dot)
h1.xplore	Element.class — NOT xplore.h1
External CSS location	In <head> section — not body
background-image	background-image: url('img.jpg')
display: block	Causes a line break between elements

Property / Concept	Key Fact
visibility:hidden vs display:none	hidden keeps space; none removes it
float on paragraph	float is NOT applicable to a paragraph
background-render	NOT a valid CSS property
border: 2px solid DodgerBlue	Correct; 'border-line' is wrong
body { background-color: yellow }	Correct CSS — sets page background
table border in CSS	Use 'border' property

Most Asked JavaScript Facts

Fact	TCS IPA Answer
x === y means	Equal in value AND data type
Division by zero	NOT an error — returns Infinity
true/false are	Nonnumeric boolean values (not 0/1)
Date stored as	Milliseconds since Jan 1, 1970
Array definition	[] list form (also valid: new Array())
new keyword for array	FALSE that it cannot be used
Different tags sharing class	Allowed — FALSE that they cannot
<div> is	Block-level element
para.firstChild.nodeValue	Correct way to change paragraph text
getElementByClass()	INVALID — use getElementsByClassName()
Invalid escape sequence	/e (valid: /n /r /t /b /f)
Server-side JS platform	LiveWire (by Netscape)
JS runs via	Script engine
NaN * 3	NaN (any operation with NaN = NaN)

Complete HTML Summary Table

All tags at a glance

Tag	Purpose	Syntax	Exam Importance
<!DOCTYPE html>	Declares HTML5	<!DOCTYPE html>	High
<html>	Root element	<html lang='en'>	High
<head>	Metadata container	<head>...</head>	High
<body>	Visible content	<body>...</body>	High
<title>	Browser tab title	<title>My Page</title>	Medium
<meta>	Metadata (charset, viewport)	<meta charset='UTF-8'>	High
<link>	External CSS	<link rel='stylesheet' href='s.css'>	High
<style>	Internal CSS	<style>p{color:red}</style>	High
<script>	JavaScript	<script type='text/javascript'>	High
<h1>—<h6>	Headings	Title	High
<p>	Paragraph	Text	High
 	Line break (void)	 	High
<hr>	Horizontal rule (void)	<hr>	Medium
	Bold (presentational)	text	High
	Bold (semantic)	important	High
<i>	Italic (presentational)	<i>text</i>	High
	Italic (emphasis)	stressed	High
<u>	Underline	<u>text</u>	Medium
<ins>	Inserted text (underlined)	<ins>text</ins>	High
<s>	Strikethrough	<s>text</s>	Medium
	Deleted text	text	Medium
<mark>	Highlighted text	<mark>text</mark>	Medium
<sup>	Superscript	x²	Medium
<sub>	Subscript	H₂O	Medium
<pre>	Preformatted text	<pre>code</pre>	High
<code>	Inline code	<code>var x</code>	Medium

Tag	Purpose	Syntax	Exam Importance
<blockquote>	Long quotation	<blockquote>text</blockquote>	High
<q>	Short inline quote	<q>text</q>	Medium
<marquee>	Scrolling text	<marquee>text</marquee>	High
<a>	Hyperlink	text	High
	Image (void)		High
	Unordered list	item	High
	Ordered list	<ol type='l'> item	High
	List item	item	High
<dl><dt><dd>	Definition list	<dl><dt>T</dt><dd>D</dd></dl>	Medium
<table>	Table	<table border='1'>	High
<tr>	Table row	<tr>...</tr>	High
<th>	Header cell	<th>Header</th>	High
<td>	Data cell	<td>Data</td>	High
<caption>	Table caption	<caption>Title</caption>	High
<form>	Form container	<form action="" method='POST'>	High
<input>	Input field (void)	<input type='text' name='n'>	High
<textarea>	Multi-line input	<textarea rows='4'>	High
<select>	Dropdown	<select><option>X</option></select>	High
<option>	Dropdown option	<option value='v'>Label</option>	High
<button>	Button	<button type='submit'>Click</button>	High
<label>	Input label	<label for='id'>Name</label>	High
<audio>	Audio player	<audio controls src='s.mp3'>	High
<video>	Video player	<video controls src='v.mp4'>	High
<source>	Media source	<source src='f.mp4' type='video/mp4'>	Medium
<iframe>	Inline frame	<iframe src='url'></iframe>	Medium
<canvas>	Scripted drawing	<canvas id='c' width='200'>	High
<svg>	Vector graphics	<svg viewBox='0 0 100 100'>	High
<header>	Page header	<header>logo nav</header>	High
<nav>	Navigation	<nav>links</nav>	High

Tag	Purpose	Syntax	Exam Importance
<main>	Main content	<main>content</main>	High
<section>	Thematic section	<section>...</section>	High
<article>	Standalone article	<article>post</article>	High
<aside>	Sidebar	<aside>related</aside>	High
<footer>	Page footer	<footer>copyright</footer>	High
<figure>	Media figure	<figure>...</figure>	Medium
<figcaption>	Figure caption	<figcaption>desc</figcaption>	Medium
<details>	Expandable widget	<details><summary>Q</summary>A</details>	Medium
<progress>	Progress bar	<progress value='70' max='100'>	Medium
	Inline container	text	High
<div>	Block container	<div class='box'>content</div>	High

Complete CSS Summary Table

All key properties

Property	Purpose	Syntax / Values	Exam Importance
color	Text colour	color: red / #FF0000 / rgb(255,0,0)	High
background-color	Element background	background-color: yellow	High
background-image	Background image	background-image: url('bg.jpg')	High
background-repeat	Repeat mode	no-repeat / repeat-x / repeat-y	Medium
background-attachment	Scroll or fixed	scroll / fixed	Medium
background-size	Image size	cover / contain / px/%	Medium
border	Shorthand border	border: 2px solid red	High
border-radius	Rounded corners	border-radius: 10px	Medium
outline	Outside border	outline: 2px solid blue	Low
margin	Outside spacing	margin: 10px / margin: top right bottom left	High
padding	Inside spacing	padding: 10px	High
width / height	Element size	width: 300px / 50%	High
max-width / min-width	Size constraints	max-width: 800px	Medium
box-sizing	Box model mode	border-box / content-box	Medium
display	Display mode	block / inline / inline-block / none / flex	High
visibility	Show/hide	visible / hidden	High
position	Positioning mode	static / relative / absolute / fixed / sticky	High
top / right / bottom / left	Offset (with position)	top: 10px	Medium
z-index	Stack order	z-index: 10	Medium
float	Float element	float: left / right / none	Medium
clear	Clear float	clear: both / left / right	Medium
overflow	Handle overflow	visible / hidden / scroll / auto	Medium
font-family	Typeface	font-family: Arial, sans-serif	High
font-size	Text size	font-size: 16px / 1em / 200%	High
font-weight	Boldness	bold / normal / 100–900	High
font-style	Italic	normal / italic / oblique	High
text-align	Alignment	left / center / right / justify	High

Property	Purpose	Syntax / Values	Exam Importance
text-decoration	Decoration	none / underline / line-through	High
text-transform	Case	uppercase / lowercase / capitalize	Medium
line-height	Line spacing	line-height: 1.5	Medium
letter-spacing	Char spacing	letter-spacing: 2px	Low
list-style-type	List bullet	disc / circle / square / none	High
border-collapse	Table border	collapse / separate	High
flex-direction	Flex direction	row / column	Medium
justify-content	Main axis align	flex-start / center / space-between	Medium
align-items	Cross axis align	stretch / center / flex-start	Medium
opacity	Transparency	opacity: 0.5	Medium
cursor	Cursor style	cursor: pointer / default	Low
transition	Animation transition	transition: all 0.3s ease	Low

Complete JavaScript Summary Table

Methods, purpose, output

Method / Concept	Purpose	Syntax / Output	Exam Importance
getElementById()	DOM select by id	document.getElementById('id')	High
getElementsByClassName()	DOM select by class	document.getElementsByClassName('c')	High
querySelector()	First CSS selector match	document.querySelector('.cls')	High
querySelectorAll()	All CSS selector matches	document.querySelectorAll('p')	High
.innerHTML	Get/set HTML inside element	el.innerHTML = 'Hi'	High
.innerText	Get/set visible text	el.innerText = 'Hello'	High
.textContent	Get/set all text	el.textContent	Medium
.value	Input field value	input.value	High
.style.prop	Inline style	el.style.color = 'red'	High
addEventListener()	Attach event	el.addEventListener('click', fn)	High
alert() / confirm() / prompt()	Browser dialogs	alert('msg')	High
typeof	Type of value	typeof 5 → 'number'	High
parseInt() / parseFloat()	Parse string to number	parseInt('5px') → 5	High
isNaN()	Check if NaN	isNaN(NaN) → true	High
eval()	Evaluate string as JS	eval('5+3') → 8	High
String.length	String length	'hello'.length → 5	High
toUpperCase() / toLowerCase()	Case conversion	'hi'.toUpperCase() → 'HI'	High
indexOf()	Find position	'hello'.indexOf('l') → 2	High
slice()	Extract substring	'hello'.slice(1,3) → 'el'	High
replace()	Replace in string	'hi'.replace('h','H') → 'Hi'	High
split()	String to array	'a,b'.split(',') → ['a','b']	High
trim()	Remove whitespace	' hi '.trim() → 'hi'	Medium
Array push() / pop()	Add/remove end	[1,2].push(3) → [1,2,3]	High
Array forEach()	Iterate	arr.forEach(fn)	High
Array map()	Transform	arr.map(x=>x*2)	High

Method / Concept	Purpose	Syntax / Output	Exam Importance
Array filter()	Filter	arr.filter(x=>x>2)	High
Math.round/floor/ceil	Rounding	Math.round(4.6) → 5	High
Math.max() / Math.min()	Max / Min	Math.max(3,7) → 7	High
Math.random()	Random 0 to <1	Math.random()	Medium
new Date()	Date object	new Date()	High
getTime()	Milliseconds since 1970	d.getTime()	High
getFullYear() / getMonth()	Date parts	d.getFullYear() → 2025	High
setTimeout()	Delay execution	setTimeout(fn, 1000)	Medium
setInterval()	Repeat execution	setInterval(fn, 1000)	Medium
JSON.parse()	String to JS object	JSON.parse(str)	Medium
JSON.stringify()	JS object to string	JSON.stringify(obj)	Medium
try / catch / finally	Error handling	try{}catch(e){}finally{}	High
throw	Throw error	throw new Error('msg')	Medium

Comparison Tables

The pairs most often confused — study these until instant

id vs class

Aspect	id	class
Selector	#name	#name vs .name → # is id, . is class
Uniqueness	Unique — one element per page	Reusable — many elements can share it
JS access	getElementById('name')	getElementsByClassName('name')
Priority	Higher specificity	Lower specificity

div vs span

Aspect	<div>	
Type	Block-level element	Inline element
Line break	Yes — starts on new line	No — flows with text
Use case	Structural section container	Style part of text inline
Width/height	Accepts width/height	Ignored by default

GET vs POST (form method)

Aspect	GET	POST
Data location	Appended to URL as query string	In HTTP request body (not visible in URL)
Security	Less secure (visible)	More secure for sensitive data
Data limit	Limited by URL length	No practical size limit
Typical use	Search, navigation	Login, file upload, form submission

inline vs block

Aspect	inline	block
Starts new line	No — flows with content	Yes — takes full width
Width/height	Cannot set explicitly	Can set explicitly
Examples	<a>	<div><p><h1> <table>

Inline CSS vs Internal CSS vs External CSS

Aspect	Inline	Internal / External
Where	style='...' on the element	<style> in head / linked .css file
Priority	Highest	Internal ~ External (cascade order)
Reusable	No — one element only	Yes — whole page / whole site
Recommended	No (maintenance)	External (best practice)

== vs ===

Aspect	== (loose equality)	=== (strict equality)
Checks	Value only (with type coercion)	Value AND type
5 == '5'	true	false
null == undefined	true	false
Use case	Avoid — causes bugs	Always prefer ===

let vs var vs const

Aspect	var	let / const
Scope	Function or global	Block scoped
Re-declare	Allowed	Not allowed
Hoisting	Yes (as undefined)	TDZ — not accessible before declaration
Re-assign	Yes	let: yes / const: NO

null vs undefined

Aspect	null	undefined
Meaning	Intentional empty value	Variable declared but not assigned
typeof	'object' (historical quirk)	'undefined'
Set by	Developer explicitly	JavaScript engine
== comparison	null == undefined → true	undefined == null → true

display:none vs visibility:hidden

Aspect	display: none	visibility: hidden
Element removed from layout	Yes — no space taken	No — invisible but space remains
Children	Also hidden and removed	Also hidden but space remains
TCS IPA ask	'Completely remove' → none	'Hide but keep space' → hidden

margin vs padding

Aspect	margin	padding
Location	OUTSIDE the border	INSIDE the border (around content)
Creates space	Between elements	Between content and border

Aspect	margin	padding
Background shows	No (transparent)	Yes — background-color shows in padding
Auto centering	margin: 0 auto centres block	padding: 0 auto is not valid for centering

position: relative vs absolute

Aspect	relative	absolute
Reference point	Its own normal position	Nearest positioned ancestor (or viewport)
Space in layout	Still occupies space	Removed from normal flow
Use case	Fine offset from normal position	Overlay, popup, tooltip

position: absolute vs fixed

Aspect	absolute	fixed
Reference	Nearest positioned ancestor	Viewport (browser window)
Scrolls with page	Yes	No — stays fixed on screen
Use case	Element relative to parent	Sticky header, floating button

em vs rem

Aspect	em	rem
Relative to	Parent element's font size	Root (<html>) element's font size
Compounding	Yes — nested em multiplies	No — always same base
Predictability	Less predictable in nesting	More predictable

strong vs b | em vs i

Aspect	 / 	 / <i>
Type	Semantic (meaning)	Presentational (appearance only)
strong	Semantically important	 is just visually bold
em	Emphasised / stressed	<i> is just visually italic
Screen readers	Convey importance	Ignored semantically

article vs section

Aspect	<article>	<section>
Self-contained	Yes — standalone content	No — needs surrounding context
Examples	Blog post, news item, comment	Chapter of a book, tabbed panel
Syndication	Can be syndicated independently	Cannot stand alone

querySelector vs getElementById

Aspect	querySelector()	getElementById()
Selector	Any CSS selector string	Only element id (no # needed)
Returns	First matching element	Element with that id (null if not found)
Multiple	querySelectorAll() for all	Not applicable
Flexibility	More flexible	Faster for id lookup

innerHTML vs innerText vs textContent

Aspect	innerHTML	innerText / textContent
Parses HTML	Yes — renders tags	innerText: no; textContent: no
Hidden elements	Includes	innerText: excludes hidden; textContent: includes
XSS risk	Yes — use carefully	No — text is not parsed

Attempt each question before reading the boxed answer. Mix of HTML, CSS, JavaScript, code-output and tricky pattern questions.

1. What is the correct syntax to set border colours in CSS?

- A. `<h1 style='border-line: 2px solid DodgerBlue;'>Hello</h1>`
- B. `<h1 style='border: solid DodgerBlue;'>Hello</h1>`
- C. `<h1 style='border: 2px solid DodgerBlue;'>Hello</h1>`
- D. None of the above

✓ Answer: C — The correct shorthand is: width style color. 'border-line' is not a valid CSS property.

2. What is the usage of `<span>` in HTML?

- A. Container for styling block-level elements
- B. Used to group inline elements in a document
- C. Used to create line breaks

✓ Answer: B — `<span>` is an inline container used to style inline content. It does not cause a line break.

3. What is the usage of `<svg>` in HTML?

- A. Adds audio controls like play and pause
- B. Draws graphics on the fly via scripting (canvas does this)
- C. Used for drawing paths, boxes and graphic vector images

✓ Answer: C — `<svg>` is for vector graphics (paths, shapes). `<canvas>` draws raster graphics via JavaScript scripting.

4. Which tag is used to insert a line break in HTML?

- A. `<br>`
- B. `<a>`
- C. `<pre>`
- D. `<b>`

✓ Answer: A — `<br>` is the line break tag. It is a void element — no closing tag needed.

5. Which character represents the closing of a tag in HTML?

- A. \
- B. !
- C. /
- D. .

✓ Answer: C — / (forward slash) closes a tag. `</p>` is the closing paragraph tag.

6. How do you create a hyperlink in HTML?

- A. `<a href='url'>text</a>`
- B. `<a url='url'>text</a>`
- C. `<a link='url'>text</a>`
- D. `<link href='url'>text</link>`

✓ **Answer: A** — `href` is the correct attribute for hyperlinks. `url` and `link` are not valid attributes.

7. How do you insert an image in HTML?

- A. `<img href='jtp.png'>`
- B. `<img url='jtp.png'>`
- C. `<img link='jtp.png'>`
- D. `<img src='jtp.png'>`

✓ **Answer: D** — Images use the `src` attribute, not `href` or `url`.

8. Which tag defines options in a dropdown selection list?

- A. `<select>`
- B. `<list>`
- C. `<dropdown>`
- D. `<option>`

✓ **Answer: D** — `<option>` defines each item inside a `<select>` dropdown.

9. Which tag creates a combo box (dropdown box)?

- A. `<list>`
- B. `<select>`
- C. `<input type='dropdown'>`
- D. `<ul>`

✓ **Answer: B** — `<select>` creates the dropdown container. `<option>` goes inside it.

10. Which attribute provides a unique name to an element?

- A. `class`
- B. `id`
- C. `type`
- D. `None`

✓ **Answer: B** — `id` gives a unique identifier. `class` can be shared by multiple elements.

11. Which HTML tag displays text with a scrolling effect?

- A. `<marquee>`
- B. `<scroll>`
- C. `<div>`
- D. `None`

✓ **Answer: A** — `<marquee>` creates scrolling text. It is deprecated in modern HTML but tested in TCS IPA.

12. Which HTML tag is a special formatting tag that preserves whitespace?

- A. `<p>`
- B. `<b>`
- C. `<pre>`
- D. None

✓ **Answer: C** — `<pre>` preserves spaces, tabs and line breaks exactly as written in the HTML.

13. How do you insert a background image in HTML (inline)?

- A. `<body background='img.png'>`
- B. `<img background='img.png'>`
- C. `<bg-image='img.png'>`
- D. None

✓ **Answer: A** — The background attribute on `<body>` sets a background image. In modern CSS: `background-image: url()`.

14. What is the smallest heading in HTML by default?

- A. H1
- B. H2
- C. H6
- D. H4

✓ **Answer: C** — H1 is the largest/most important; H6 is the smallest.

15. What is the correct syntax for an HTML comment?

- A. `<!-- Comment -->`
- B. `// Comment`
- C. `# Comment`
- D. `/* Comment */`

✓ **Answer: A** — HTML comments use `<!-- -->`. CSS uses `/* */`. JS uses `//` or `/* */`.

16. Which HTML element defines description data in a definition list?

- A. `<li>`
- B. `<ol>`
- C. `<dd>`
- D. `<dl>`

✓ **Answer: C** — `<dd>` is the definition description. `<dt>` is the term. `<dl>` is the container.

17. Do all HTML tags have closing tags?

- A. True

B. False

✓ **Answer: B** — Void elements like `<br>`, `<img>`, `<input>`, `<hr>`, `<meta>` have no closing tag.

18. What is the command for an ordered list with Roman numerals?

A. `<ol type='R'>`

B. `<ol type='I'>`

C. `<ol type='roman'>`

D. `<ol type='num'>`

✓ **Answer: B** — `type='I'` gives uppercase Roman numerals. `type='i'` gives lowercase.

19. Which is an invalid HTML5 field type?

A. date

B. Datetime

C. month

D. tel

✓ **Answer: B** — Datetime is not valid. The correct type is `datetime-local`. date, month, tel are all valid.

20. How are quotations defined in HTML?

A. `<quote>`

B. `<block>`

C. `<blockquote>`

D. None

✓ **Answer: C** — `<blockquote>` is for long quotations. `<q>` is for short inline quotes.

21. What are the types of unordered list bullets in HTML?

A. Circle, square, disc

B. Triangle, square, disc

C. Triangle, circle, disc

D. All of the above

✓ **Answer: A** — Valid list-style-type values for `<ul>`: disc (default), circle, square. Triangle is NOT valid.

22. Which tag is used to emphasise text?

A. `<b>`

B. `<strong>`

C. `<em>`

✓ **Answer: C** — `<em>` emphasises text (semantic). `<strong>` makes it importantly bold. `<b>` is just bold.

23. Which tag is NOT a formatting tag?

A. `<b>`

B. `<p>`

- C.
- D. <i>

✓ **Answer: B** — <p> defines a paragraph, not a text-formatting tag.

24. The <caption> tag is used with:

- A. Image
- B. Table
- C. Both
- D. None

✓ **Answer: B** — <caption> adds a title to a <table>. It is not used with images.

25. Which of these is NOT inline in HTML?

- A. <hr>
- B.

- C.
- D. <a>

✓ **Answer: A** — <hr> is a block-level element.
, , <a> are inline elements.

26. Which is NOT a valid colour in HTML?

- A. Color: darkblue
- B. Color: #oooff
- C. Color: rgb(0,0,255)
- D. Color: blue

✓ **Answer: B** — #oooff uses the letter 'o' instead of zero '0'. Hex colours must use 0-9 and A-F only.

27. What appears when opacity='0.25' is set as an HTML attribute on ?

- A. Display with normal appearance
- B. Display image as watermark (semi-transparent)
- C. Image is hidden

✓ **Answer: B** — Setting opacity via inline style makes the image semi-transparent (watermark-like).

28. What is the appearance of the <ins> tag?

- A. Italics
- B. Underline
- C. Bold
- D. Unformatted

✓ **Answer: B** — <ins> marks inserted text and renders with an underline by default.

29. Can a link be an image or any other HTML element?

- A. True

B. False

✓ **Answer: A** — Any HTML element can be wrapped in an `<a>` tag to make it a hyperlink.

30. Which property is NOT applicable for a paragraph?

A. float

B. border

C. margin

D. font-family

✓ **Answer: A** — Per TCS IPA, float is not applicable to a paragraph. border, margin, font-family are valid.

31. Which block of code will successfully apply a CSS comment?

A. `p {<!-- comment --> color: red; }`

B. `p {/* comment */ color: red; }`

C. `p {// comment color: red; }`

D. `p {# comment color: red; }`

✓ **Answer: B** — CSS comments use `/* */`. HTML-style `<!-- -->` and JS-style `//` are invalid in CSS.

32. Which block styles the `<h1>` with class 'Xplore'? `<h1 class='Xplore'>`

A. `h1 {...}`

B. `xplore.h1 {...}`

C. `h1 {...}`

D. `h1.xplore {...}`

✓ **Answer: D** — element.class selector: `h1.xplore` — element first, then class name with dot, lowercase.

33. How can we select an element with a specific class in CSS?

A. `#`

B. `.`

C. `<`

D. None

✓ **Answer: B** — Class selector uses `.` (dot). ID selector uses `#` (hash).

34. How do we write comments in CSS?

A. `/* */`

B. `//`

C. `#`

D. All

✓ **Answer: A** — Only `/* */` is valid in CSS for comments.

35. What does CSS stand for?

A. Cascading Style Sheets

- B. Common Style Sheets
- C. Central Style Sheets
- D. Control Style Sheets

✓ **Answer: A — CSS = Cascading Style Sheets. The cascade refers to rule priority/inheritance.**

36. How do you define an ID selector in CSS?

- A. # (hash)
- B. @ (at)
- C. \$ (dollar)
- D. None

✓ **Answer: A — ID selectors use the # symbol: #myId { color: red; }**

37. Where should an external stylesheet be referenced in HTML?

- A. In the <head> section
- B. At the end of the document
- C. At the beginning
- D. In <body>

✓ **Answer: A — External CSS is linked in the <head> section using <link rel='stylesheet' href='file.css'>.**

38. Which is not a CSS unit?

- A. em
- B. ut
- C. %
- D. px

✓ **Answer: B — 'ut' is not a CSS unit. Valid units include px, em, rem, %, vh, vw, cm, mm.**

39. What is the syntax for background-image in CSS?

- A. background-image: url('file.jpg')
- B. bg-image: 'file.jpg'
- C. background: file.jpg
- D. image-background: url()

✓ **Answer: A — background-image: url('path') is the correct syntax.**

40. What does display: block result in between two elements?

- A. Two-tab spacings
- B. Two-line breaks
- C. A line break
- D. A tab spacing

✓ **Answer: C — display: block causes a single line break — each block element starts on a new line.**

41. Which CSS syntax will HIDE an element completely?

- A. `h1.hidden {visibility: hidden;}`
- B. `h1.hidden {display: none;}`
- C. `h1.hidden {display: hidden;}`
- D. `h1.hidden {visibility: none;}`

✓ **Answer: B** — `display:none` removes the element from layout entirely. `visibility:hidden` keeps the space.

42. Which of the following correctly sets font size?

- A. `h2 {font-size: 200pixels;}`
- B. `h2 {font-size: 200;}`
- C. `h2 {font-size: 200px;}`
- D. `h2 {font-size: 200%;}`

✓ **Answer: C and D** — Both `200px` and `200%` are valid. `'200pixels'` is invalid. Bare `'200'` without a unit is also invalid.

43. Which is NOT a valid CSS property?

- A. `background-render`
- B. `background-attachment`

✓ **Answer: A** — `background-render` does not exist. `background-attachment (scroll/fixed)` is valid.

44. Given `<p class='A'>ABC</p>`, how do you style 'ABC' in CSS?

- A. `A {color: red;}`
- B. `#A {color: red;}`
- C. `*A {color: red;}`
- D. `.A {color: red;}`

✓ **Answer: D** — `.A` selects elements with `class='A'`. `#A` would select an element with `id='A'`.

45. Which HTML sequence is correct for the document structure?

- A. `<html><head><title></title><body></body></head></html>`
- B. `<html><title></title><head></head><body></body></html>`
- C. `<html><head></head><body><title></title></body></html>`
- D. `<html><head><title></title></head><body></body></html>`

✓ **Answer: D** — Correct order: `html > head > title (inside head)`, then `body` after `head`.

46. To specify table border in CSS, which property is used?

- A. `table-border`
- B. `border`
- C. `tb-border`
- D. `tbl-border`

✓ **Answer: B** — The standard 'border' property applies to tables just like any other element.

47. In JavaScript, which of the following is NOT considered an error?

- A. Syntax error
- B. Missing semicolons
- C. Division by zero
- D. Missing bracket

✓ **Answer: C** — Division by zero returns Infinity — it is NOT an error in JavaScript.

48. In JavaScript, x === y means:

- A. Equal in value, type and reference address
- B. Equal in value only
- C. Equal in value and data type
- D. Not the same at all

✓ **Answer: C** — === is strict equality: checks value AND data type. It does not check reference address.

49. Which snippet checks if variable 'a' is NOT equal to null?

- A. if(a!==null)
- B. if(a!)
- C. if(a!null)
- D. if(a!=null)

✓ **Answer: A** — a !== null uses strict inequality. a != null would also work loosely, but A is the correct TCS answer.

50. Which of the following colours contain equal amounts of RGB?

- A. White
- B. Gray
- C. Black
- D. All of the above

✓ **Answer: D** — White (255,255,255), Gray (128,128,128) and Black (0,0,0) all have equal R, G, B values.

51. Given: var x; x = myFunction(NaN, 3); function myFunction(a,b){return a*b;} — what is x?

- A. 3
- B. NaN
- C. 0
- D. undefined

✓ **Answer: B** — Any arithmetic with NaN returns NaN. NaN * 3 = NaN.

52. Which is an invalid escape sequence in JavaScript?

- A. /e

- B. /r
- C. /f
- D. /b

✓ Answer: A — Valid: \n \r \t \b \f \v \0. \e does not exist in JavaScript.

53. JavaScript runs using a:

- A. Compiler
- B. Script engine
- C. Interpreter only

✓ Answer: B — JavaScript runs via a script engine (e.g. V8 in Chrome). The statement 'JS runs using a script engine' is TRUE.

54. The following code: `<p class='abc'></p> <script>var x; document.getElementsByClassName('abc').innerHTML=x;</script>` — output?

- A. NaN
- B. 0
- C. 1
- D. Error

✓ Answer: D — `getElementsByClass()` does not exist. The correct method is `getElementsByName()`. This throws a `TypeError`.

55. Which statement is correct about JavaScript booleans?

- A. true = 1 and false = 0
- B. true = 0 and false = 1
- C. true is any nonzero value
- D. true and false are nonnumeric values

✓ Answer: D — In JS, true and false are boolean type — not numeric. `typeof true` → 'boolean', not 'number'.

56. Given: `<h2 class='fruit mango'>I love fruits</h2>` — which is true?

- A. Invalid HTML statement
- B. Heading with class named 'fruit mango'
- C. Heading with two classes: 'fruit' and 'mango'
- D. None

✓ Answer: C — Multiple classes are space-separated in the class attribute. This element has two classes.

57. Can different tags share the same class?

- A. TRUE — they cannot share
- B. FALSE — they can share

✓ Answer: B — Multiple different elements (p, div, span etc.) can all have the same class name.

58. Is <div> a block-level element?

- A. TRUE
- B. FALSE

✓ **Answer: A** — <div> is a block-level element. is the inline equivalent.

59. Which correctly creates and initialises a string variable in JavaScript?

- A. var str1=Hello;
- B. var str1='Hello';
- C. var str1="Hello";
- D. var str1="Hello";

✓ **Answer: B** — String values must be in quotes. 'Hello' and "Hello" are both valid.

60. Which HTML elements were added in HTML5?

- A. <main>
- B. <header>
- C. <article>
- D. <footer>
- E. All of them

✓ **Answer: E** — All of: <main>, <header>, <footer>, <article>, <section>, <aside>, <nav>, <canvas>, <svg> were added in HTML5.

61. Which option opens Google in a new window?

- A. target='_top'
- B. target='_self'
- C. target='_parent'
- D. target='_blank'

✓ **Answer: D** — _blank opens the link in a new tab/window. _self (default) opens in the same tab.

62. How does JavaScript store dates in a Date object?

- A. Seconds since Netscape's IPO
- B. Milliseconds since January 1st, 1970
- C. Days since January 1st, 1900
- D. None

✓ **Answer: B** — JavaScript Date = milliseconds since the Unix Epoch (Jan 1, 1970 UTC).

63. What is the result of: var res = eval('x + y') where x=12, y=8?

- A. 'x+y'
- B. 20
- C. NaN
- D. Error

✓ **Answer: B** — `eval()` evaluates the string as JavaScript code. $x+y = 12+8 = 20$.

64. What is the output of: `let sum=0; const a=[1,2,3]; a.forEach(getSum); function getSum(ele){sum+=ele;} — print(sum)`?

- A. 0
- B. 1
- C. 6
- D. undefined

✓ **Answer: C** — `forEach` iterates all elements. $sum = 0+1+2+3 = 6$. The function modifies the outer `sum` variable.

65. Which tag would you use to insert JavaScript in an HTML page?

- A. `<script type='text/javascript'>`
- B. `<HTMLScript>`
- C. `<JScript>`
- D. `<JavaScript>`

✓ **Answer: A** — The correct tag is `<script type='text/javascript'>`. All other options are invalid.

66. Server-side JavaScript is known as:

- A. Navigator
- B. Microsoft
- C. LiveWire
- D. Native

✓ **Answer: C** — LiveWire was Netscape's server-side JavaScript platform.

67. If `para1` is a DOM paragraph object, how do you change its text?

- A. `para1.firstChild.nodeValue = 'New Text'`
- B. `para1.nodeValue = 'New Text'`
- C. `para1 = 'New Text'`
- D. `para1.value = 'New Text'`

✓ **Answer: A** — Text is in the first child text node. Access it via `firstChild.nodeValue`.

68. Which CSS hides an element while keeping its space in the layout?

- A. `display: none`
- B. `visibility: hidden`
- C. `display: hidden`
- D. `visibility: none`

✓ **Answer: B** — `visibility: hidden` keeps the space. `display: none` removes it entirely.

69. What will `100 + 5 ^ 3` produce in JavaScript per TCS IPA answer?

- A. 125

B. 225

C. NaN

✓ **Answer: B** — Per TCS IPA PDF answer: 225. (Note: in real JS, ^ is XOR and result would differ — but follow the TCS PDF here.)

70. Arrays in JavaScript are defined using:

A. List form []

B. Tuple form ()

C. Dictionary form {}

D. None

✓ **Answer: A** — Arrays use [] — var a = [1,2,3]. Objects use {}.

71. Given: <h1 class='Xplore'> — which CSS rule styles it?

A. h1 {...}

B. xplore.h1 {...}

C. h1 {...}

D. h1.xplore {...}

✓ **Answer: D** — element.class format: h1.xplore — element name first, then dot, then class name (lowercase).

72. Which tag is used for inserting JavaScript in HTML?

A. <script type='text/javascript'>

B. <script>

C. Both A and B

D. <js>

✓ **Answer: C** — Both forms are accepted. type='text/javascript' was required in HTML4 but is optional in HTML5.

73. What is the output of: <style>body{background-color:yellow}</style>?

A. True — sets background to yellow

B. False — invalid syntax

✓ **Answer: A** — This is valid CSS. It sets the background colour of the entire page to yellow.

74. Which correctly represents a string variable?

A. var str1=Hello;

B. var str1='Hello';

C. var str1="Hello";

D. var str1="Hello";

✓ **Answer: B** — Strings must be in matching quotes. var str1='Hello'; or var str1="Hello"; are correct.

75. Which loop runs its body AT LEAST ONCE before checking the condition?

- A. for
- B. while
- C. do...while
- D. for...of

✓ **Answer: C** — **do...while** executes the body first, then checks the condition at the end.

76. What does the window object represent in JavaScript?

- A. The browser window (global object)
- B. The HTML document
- C. The DOM tree
- D. The CSS stylesheet

✓ **Answer: A** — **window** is the global browser object. **document** is the HTML document within window.

77. Which method is used to find the first element matching a CSS selector?

- A. getElementById()
- B. querySelector()
- C. getElementsByClassName()
- D. querySelectorAll()

✓ **Answer: B** — **querySelector()** returns the first match. **querySelectorAll()** returns all matches.

78. What does innerHTML do?

- A. Gets/sets only text content
- B. Gets/sets HTML markup inside element
- C. Gets element attributes
- D. Removes element from DOM

✓ **Answer: B** — **innerHTML** parses and renders HTML tags. **innerText** only handles plain text.

79. Which is the correct way to add a click event in JavaScript?

- A. el.addEvent('click', fn)
- B. el.addEventListener('click', fn)
- C. el.onClick(fn)
- D. el.click = fn

✓ **Answer: B** — **addEventListener()** is the standard, modern way to attach events.

80. What is typeof null in JavaScript?

- A. 'null'
- B. 'undefined'
- C. 'object'
- D. 'number'

✓ **Answer: C** — **typeof null** returns 'object' — a historical bug in JavaScript that was never fixed.

81. Which method converts a JS object to a JSON string?

- A. JSON.parse()
- B. JSON.stringify()
- C. JSON.convert()
- D. JSON.toText()

✓ **Answer: B** — `JSON.stringify()` → JS object to string. `JSON.parse()` → JSON string back to object.

82. What is the result of `Math.floor(4.9)`?

- A. 5
- B. 4
- C. 4.9
- D. NaN

✓ **Answer: B** — `Math.floor()` rounds DOWN. `4.9` → `4`. Use `Math.ceil()` to round up.

83. Which statement stops a loop early?

- A. continue
- B. break
- C. stop
- D. exit

✓ **Answer: B** — `break` exits the loop immediately. `continue` skips to the next iteration.

84. What does 'use strict' do in JavaScript?

- A. Forces all variables to be typed
- B. Enables strict mode — catches silent errors
- C. Disables all functions
- D. Makes code run faster only

✓ **Answer: B** — 'use strict' enables strict mode, which throws errors for undeclared variables etc.

85. Which is NOT a JavaScript data type?

- A. number
- B. string
- C. boolean
- D. character

✓ **Answer: D** — JS does not have a 'character' type. Individual characters are just single-character strings.

86. What is `isNaN('hello')`?

- A. true
- B. false
- C. NaN

D. Error

✓ **Answer: A** — `isNaN('hello')` → true because 'hello' cannot be converted to a number.

87. Which correctly accesses a CSS-styled element in JavaScript?

- A. `document.getElementsByClassName('name')`
- B. `document.getElementsByClassName('name')`
- C. `document.getByClass('name')`
- D. `document.findClass('name')`

✓ **Answer: B** — `getElementsByClassName()` — plural Elements, ClassName (not Class). `getElementsByClass` is INVALID.

88. What is the output? `var x = 5; var y = '5'; console.log(x == y); console.log(x === y);`

- A. true, true
- B. false, false
- C. true, false
- D. false, true

✓ **Answer: C** — `==` does type coercion (`5 == '5' → true`). `===` is strict (`5 === '5' → false`, different types).

89. Which method removes the last element from a JavaScript array?

- A. `push()`
- B. `pop()`
- C. `shift()`
- D. `splice()`

✓ **Answer: B** — `pop()` removes and returns the last element. `shift()` removes the first element.

90. `setTimeout(fn, 2000)` will execute fn after:

- A. 2 seconds
- B. 2 milliseconds
- C. 200 seconds
- D. Immediately

✓ **Answer: A** — Delay is in milliseconds. `2000ms = 2 seconds`.

91. Which tag defines the main content area of an HTML page?

- A. `<body>`
- B. `<content>`
- C. `<main>`
- D. `<div>`

✓ **Answer: C** — `<main>` semantically marks the primary content. There should be only one `<main>` per page.

92. What does 'relative' positioning do in CSS?

- A. Removes element from normal flow
- B. Offsets element from its normal position without removing it from flow
- C. Fixes element to the viewport
- D. Same as static

✓ **Answer: B** — **position: relative** moves the element relative to its normal position but keeps the space.

93. Which property centres a block element horizontally?

- A. text-align: center
- B. margin: 0 auto
- C. padding: auto
- D. align: center

✓ **Answer: B** — **margin: 0 auto** horizontally centres a block element (needs a defined width).

94. What does Array.filter() return?

- A. A single value
- B. A new array with elements that pass the test
- C. The original array modified
- D. A boolean

✓ **Answer: B** — **filter()** returns a new array containing only elements where the callback returns true.

95. Which is true about const in JavaScript?

- A. Can be re-declared
- B. Can be re-assigned
- C. Cannot be re-assigned after declaration
- D. Is function-scoped

✓ **Answer: C** — **const** creates a block-scoped, read-only binding. Re-assigning throws **TypeError**.

96. What is the correct way to handle errors in JavaScript?

- A. if (error) { }
- B. try { } catch(e) { }
- C. error { } handle { }
- D. check { } fix { }

✓ **Answer: B** — **try...catch** is the standard error-handling mechanism in JavaScript.

97. Which HTML5 element groups navigation links?

- A. <group>
- B. <links>
- C. <nav>
- D. <menu>

✓ **Answer: C** — **<nav>** is the semantic tag for primary navigation link groups.

98. What is the difference between <article> and <section>?

- A. No difference
- B. article is self-contained; section is a thematic grouping needing context
- C. section is self-contained; article needs context
- D. Both must appear inside <main>

✓ **Answer: B** — <article> can stand alone (blog post, news). <section> groups related content but needs the surrounding page.

99. Which selector has the highest CSS specificity?

- A. Element selector (p)
- B. Class selector (.cls)
- C. ID selector (#id)
- D. Universal selector (*)

✓ **Answer: C** — Specificity order: inline > ID > class > element > universal.

100. What is the output of: for(var i=0; i<3; i++) { console.log(i); }?

- A. 0 1 2
- B. 1 2 3
- C. 0 1 2 3
- D. undefined

✓ **Answer: A** — i starts at 0, loop runs while i < 3, so outputs 0, 1, 2.

101. What does the for...in loop iterate over in JavaScript?

- A. Values of an array
- B. Keys/properties of an object
- C. DOM nodes
- D. Characters of a string

✓ **Answer: B** — for...in iterates over enumerable property keys of an object. Use for...of for array values.

102. Which tag is self-closing (void) in HTML?

- A. <p>
- B. <div>
- C.
- D.

✓ **Answer: C** — Void elements have no closing tag: ,
, <hr>, <input>, <meta>, <link>.

103. What is the default value of the method attribute in an HTML form?

- A. POST
- B. GET
- C. PUT

D. None

✓ **Answer: B — If method is not specified, forms default to GET.**

104. Which CSS property controls the stacking order of elements?

A. position

B. z-index

C. stack

D. order

✓ **Answer: B — z-index controls the stacking order. Higher values appear on top. Requires position other than static.**

105. What is hoisting in JavaScript?

A. Moving code to a server

B. Variable/function declarations are moved to top of their scope before execution

C. Linking CSS and JS files

D. None

✓ **Answer: B — var declarations and function declarations are hoisted. let and const are in the TDZ.**

106. Which attribute makes an input field required before form submission?

A. mandatory

B. required

C. validate

D. must

✓ **Answer: B — required is the HTML5 attribute that prevents form submission if the field is empty.**

107. What is the difference between == and === in JS?

A. No difference

B. == checks value only (coercion); === checks value and type (strict)

C. === checks reference; == checks value

D. == is for objects; === is for primitives

✓ **Answer: B — === is the strict equality operator — both value and type must match.**

108. Which event fires when a user changes an input field?

A. onsubmit

B. onclick

C. onchange

D. oninput

✓ **Answer: C — onchange fires when the value changes and element loses focus. oninput fires on every keystroke.**

109. What is the CSS box model order from outside to inside?

- A. content → padding → border → margin
- B. margin → border → padding → content
- C. padding → margin → border → content
- D. border → margin → padding → content

✓ **Answer: B — Outside to inside: margin → border → padding → content.**

110. Which method converts a string to an integer in JavaScript?

- A. toInt()
- B. Integer.parse()
- C. parseInt()
- D. numberFrom()

✓ **Answer: C — parseInt() parses a string and returns an integer. parseFloat() for decimals.**

111. What is the output of: typeof undefined?

- A. 'null'
- B. 'undefined'
- C. 'object'
- D. 'void'

✓ **Answer: B — typeof undefined → 'undefined'. Contrast: typeof null → 'object'.**

112. Which CSS property makes text all capital letters?

- A. font-transform: uppercase
- B. text-transform: uppercase
- C. font-style: uppercase
- D. text-style: capitalize

✓ **Answer: B — text-transform: uppercase converts all text to capitals without changing the HTML.**

113. What does position: fixed do?

- A. Fixes element relative to parent
- B. Fixes element to viewport — does not scroll with page
- C. Same as position: absolute
- D. Fixes element to its normal position

✓ **Answer: B — Fixed elements stay at the same position in the viewport even when the page scrolls.**

114. Which JS method executes a function repeatedly at a set interval?

- A. setTimeout()
- B. setInterval()
- C. setRepeat()
- D. loop()

✓ **Answer: B — setInterval(fn, ms) repeats fn every ms milliseconds. clearInterval() stops it.**

115. What is the result of: 'hello'.indexOf('l')?

- A. 1
- B. 2
- C. 3
- D. -1

✓ **Answer: B** — **indexOf** returns the first position (0-based). 'h'=0,'e'=1,'l'=2 → returns 2.

116. Which is the correct way to declare a JavaScript function?

- A. function = myFunc() { }
- B. function myFunc() { }
- C. def myFunc() { }
- D. void myFunc() { }

✓ **Answer: B** — **function** keyword followed by name and parentheses. **def** is Python; **void** is Java.

117. What does border-collapse: collapse do in CSS tables?

- A. Removes the table border
- B. Merges adjacent cell borders into a single border
- C. Adds extra spacing between cells
- D. Makes table invisible

✓ **Answer: B** — **border-collapse: collapse** merges adjacent cell borders, making the table look cleaner.

118. Which HTML attribute specifies where a form submits data?

- A. method
- B. action
- C. href
- D. target

✓ **Answer: C** — Wait — **action** specifies the URL. **href** is for links. So answer is **B: action**.

119. What does Math.random() return?

- A. A random integer between 0 and 10
- B. A random float between 0 (inclusive) and 1 (exclusive)
- C. A random integer between 1 and 100
- D. A random boolean

✓ **Answer: B** — **Math.random()** → float in [0, 1). Multiply and floor to get integers.

120. What is the purpose of the <label> tag?

- A. Creates a heading
- B. Associates a text description with an input element
- C. Creates a list
- D. Defines a hyperlink

✓ **Answer: B** — `<label for='inputId'>` associates text with an input, improving accessibility and click target.

121. Which is the correct HTML5 doctype declaration?

- A. `<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN">`
- B. `<!DOCTYPE html>`
- C. `<html doctype='html5'>`
- D. `<?xml version='1.0'>`

✓ **Answer: B** — HTML5 uses the simple `<!DOCTYPE html>`. It must be the very first line.

122. What does the 'defer' attribute on a `<script>` tag do?

- A. Delays script download
- B. Executes script after HTML is fully parsed
- C. Makes script run in background
- D. Disables the script

✓ **Answer: B** — `defer` delays script execution until after the HTML document is fully parsed.

123. What is the output of: `[1,2,3].map(x => x * 2)`?

- A. `[1,2,3]`
- B. `[2,4,6]`
- C. `12`
- D. `[3,6,9]`

✓ **Answer: B** — `map()` creates a new array with each element transformed. $1*2=2$, $2*2=4$, $3*2=6$.

124. Which element creates a multi-line text input?

- A. `<input type='multiline'>`
- B. `<input type='text' rows='4'>`
- C. `<textarea>`
- D. `<textbox>`

✓ **Answer: C** — `<textarea>` is the correct element for multi-line text input.

125. What is event bubbling in JavaScript?

- A. Events fire only on the target element
- B. Events propagate from child to parent up the DOM tree
- C. Events propagate from parent to child
- D. Events cancel automatically

✓ **Answer: B** — **Bubbling:** event fires on target, then propagates up through parent elements.

126. Which CSS property adds rounded corners to an element?

- A. `corner-radius`

- B. border-radius
- C. radius
- D. round-border

✓ **Answer: B** — border-radius: 10px rounds the corners. 50% makes a perfect circle.

127. What does 'NaN === NaN' evaluate to in JavaScript?

- A. true
- B. false
- C. NaN
- D. undefined

✓ **Answer: B** — NaN is never equal to itself. Use isNaN() or Number.isNaN() to check for NaN.

128. Which HTML attribute defines an alternative text for an image?

- A. title
- B. src
- C. alt
- D. desc

✓ **Answer: C** — alt provides alternative text for screen readers and when the image fails to load.

129. What is the output of: typeof []?

- A. 'array'
- B. 'object'
- C. 'list'
- D. 'undefined'

✓ **Answer: B** — typeof [] → 'object'. Use Array.isArray([]) to specifically check for arrays.

130. Which pseudo-class styles a link that has been visited?

- A. a:hover
- B. a:active
- C. a:visited
- D. a:focus

✓ **Answer: C** — a:visited styles links the user has already clicked/visited.

131. What does the placeholder attribute do on an input?

- A. Sets the default value
- B. Shows hint text that disappears when user types
- C. Makes the field required
- D. Validates the input

✓ **Answer: B** — placeholder shows grey hint text inside the input until the user starts typing.

132. Which method returns true if ALL elements in array pass a test?

- A. filter()
- B. some()
- C. every()
- D. find()

✓ **Answer: C** — every() returns true if all elements pass. some() returns true if at least one passes.

133. What does CSS flex do?

- A. Makes text bold
- B. Enables flexible box layout on a container
- C. Sets fixed width
- D. Animates element

✓ **Answer: B** — display: flex turns the container into a flexbox, allowing flexible alignment of children.

134. What is the purpose of 'this' in JavaScript?

- A. Refers to the previous variable
- B. Refers to the current execution context (often the calling object)
- C. Refers to the next line
- D. Always refers to window

✓ **Answer: B** — 'this' refers to the object that owns the executing function. Context depends on how the function is called.

135. Which HTTP method is used by default in HTML forms?

- A. POST
- B. GET
- C. PUT
- D. DELETE

✓ **Answer: B** — The default form method is GET. It appends data to the URL as query string parameters.

136. What does Array.indexOf() return when element is not found?

- A. 0
- B. null
- C. undefined
- D. -1

✓ **Answer: D** — indexOf() returns -1 when the element is not in the array.

137. Which CSS value makes an element invisible but keeps its space?

- A. display: none
- B. opacity: 0
- C. visibility: hidden

D. Both B and C

✓ **Answer: D** — Both `opacity:0` and `visibility:hidden` hide the element but keep its space. `display:none` removes space.

138. What is a closure in JavaScript?

A. A way to close the browser

B. A function that remembers variables from its outer scope even after the outer function has returned

C. A loop that closes automatically

D. A method to end a program

✓ **Answer: B** — Closures let inner functions access the outer function's variables after the outer function has finished.

139. Which attribute makes an input field read-only?

A. disabled

B. readonly

C. locked

D. static

✓ **Answer: B** — `readonly` prevents editing but the value is submitted. `disabled` prevents editing and submission.

140. What is the output of: `console.log(1 + '2')`?

A. 3

B. '12'

C. 12

D. NaN

✓ **Answer: B** — When a number is added to a string, JS concatenates. `1 + '2' = '12'` (string).

141. Which tag wraps a side note or tangentially related content?

A. `<note>`

B. `<sidebar>`

C. `<aside>`

D. `<section>`

✓ **Answer: C** — `<aside>` semantically marks sidebar content, pull quotes, or related but non-essential content.

142. What does `Array.splice(2, 1)` do?

A. Returns element at index 2

B. Removes 1 element starting at index 2

C. Inserts at index 2

D. Copies from index 2

✓ **Answer: B** — `splice(startIndex, deleteCount)` removes `deleteCount` elements starting at `startIndex`.

143. Which CSS property controls element transparency?

- A. transparency
- B. alpha
- C. opacity
- D. filter: blur

✓ **Answer: C** — opacity: 0 is fully transparent; opacity: 1 is fully opaque.

144. What does the <details> tag do?

- A. Shows metadata
- B. Creates an expandable/collapsible widget
- C. Provides table details
- D. Defines a definition

✓ **Answer: B** — <details> wraps content that is hidden by default and revealed when clicked. <summary> is the visible heading.

145. Which method joins array elements into a string?

- A. concat()
- B. toString()
- C. join()
- D. merge()

✓ **Answer: C** — join(separator) converts array to string with the given separator. [1,2,3].join('-') → '1-2-3'.

146. What is the correct CSS selector for a paragraph inside a div?

- A. div + p
- B. div > p
- C. div p
- D. p div

✓ **Answer: C** — div p selects any <p> descending from a <div> (any depth). div > p selects only direct children.

147. Which attribute specifies that an input should be validated as an email?

- A. <input type='mail'>
- B. <input type='email'>
- C. <input validate='email'>
- D. <input type='text' format='email'>

✓ **Answer: B** — <input type='email'> automatically validates the format (must contain @).

148. What does window.location.href do?

- A. Returns the page title
- B. Returns or sets the full URL of the current page

- C. Reloads the page
- D. Returns the hostname only

✓ **Answer: B** — `window.location.href` is the full URL. Set it to navigate: `window.location.href = 'url'`.

149. Which is the correct syntax for a JavaScript arrow function?

- A. `function => (x) { return x*2; }`
- B. `(x) => x * 2`
- C. `x -> x * 2`
- D. `arrow(x) { return x*2; }`

✓ **Answer: B** — **Arrow function syntax:** `(params) => expression` or `(params) => { return value; }`

150. What does the 'async' keyword do on a function?

- A. Makes it run faster
- B. Makes the function return a Promise and allows await inside it
- C. Makes it run in a separate thread
- D. Disables error handling

✓ **Answer: B** — **async functions always return a Promise and enable use of await for asynchronous code.**

This section condenses the entire handbook. If you have read the full book, this is all you need in the final hour.

Core Concepts in One Line Each

- HTML = structure, CSS = style, JavaScript = behaviour. Always in that priority.
- CSS location: External and Internal → <head>. Inline → style attribute on element.
- CSS priority: inline > internal/external (last defined wins in cascade).
- Void tags (no closing):
 <hr> <input> <meta> <link> <source>.
- Block elements: <div> <p> <h1–h6> <table> <form> <section> <article>.
- Inline elements: <a> <i> <label> <input>.
- All HTML5 additions: <header><nav><main><section><article><aside><footer><figure><details><summary><canvas><svg><progress><meter><dialog>.
- JavaScript is NOT type-strict: typeof null = 'object', typeof undefined = 'undefined'.
- NaN * anything = NaN. Division by zero = Infinity (NOT an error). Missing semicolon = NOT an error.
- Date stored as: milliseconds since January 1, 1970.
- Array: [], Object: {}, Function: function(){} or () =>.
- Hoisting: var and functions are hoisted. let and const are NOT (TDZ).

HTML Rapid Revision

Tag	Key Fact for Exam
<marquee>	ONLY tag for scrolling text effect
 	Line break — void, NO </br>
<pre>	Special formatting — preserves all whitespace
<ins>	Appears UNDERLINED (not <u> in semantic context)
<ol type="I">	Roman numeral ordered list
<caption>	Table caption — inside <table>, NOT for images
<select>	Dropdown combo box — contains <option> tags
	Opens link in NEW tab
	src not href; alt mandatory
<dd>	Definition data in a <dl> list
<blockquote>	Long quotation — NOT <quote>

Tag	Key Fact for Exam
Datetime type	INVALID input type — use datetime-local
id attribute	Unique per page — for unique identity
class attribute	Shared by many — for grouping/styling
All HTML5 tags	Added in HTML5: main, header, footer, article, section, aside, nav, canvas, svg etc.

CSS Rapid Revision

Property / Rule	Key Fact for Exam
CSS comment	<code>/* */</code> only — NOT <code>//</code> and NOT <code>#</code>
Class selector	<code>.</code> (dot) — e.g. <code>.myClass {}</code>
ID selector	<code>#</code> (hash) — e.g. <code>#myId {}</code>
<code>h1.xplore</code>	Element.class — NOT <code>xplore.h1</code>
border shorthand	width style color — border: 2px solid red
background-render	NOT a valid property
<code>display: block</code>	Causes line break between elements
<code>display: none</code>	Removes from layout completely
<code>visibility: hidden</code>	Hides but KEEPS the space
float not for paragraph	float is NOT applicable to <code><p></code> in TCS MCQ
margin vs padding	margin = outside; padding = inside (around content)
<code>position: fixed</code>	Stays on screen — does NOT scroll with page
z-index	Stacking order — higher = on top
Link pseudo order	<code>a:link</code> → <code>a:visited</code> → <code>a:hover</code> → <code>a:active</code> (LoVeHA)
font-size: 200px / 200%	Both valid. '200pixels' and '200' (no unit) are NOT.
table border	Use 'border' property — 'table-border' is NOT valid

JavaScript Rapid Revision

Fact	TCS IPA Answer
<code>===</code> means	Equal in VALUE and DATA TYPE
<code>==</code> means	Equal in VALUE only (type coercion)
<code>a !== null</code>	Correct way to check not null
<code>NaN * 3</code>	NaN

Fact	TCS IPA Answer
Division by zero	Infinity — NOT an error
Missing semicolons	NOT an error (ASI)
true / false type	Nonnumeric boolean — NOT 0 and 1
typeof null	'object' (legacy quirk)
typeof undefined	'undefined'
Date stored as	Milliseconds since Jan 1, 1970
Arrays defined with	[] — var a = [1,2,3]
new Array() — can it?	YES — 'Cannot create with new' is FALSE
Different tags share class?	YES — 'They cannot share' is FALSE
<div> is	Block-level — TRUE
para1.firstChild.nodeValue	Correct way to change paragraph text
getElementByClass()	INVALID — use getElementsByClassName()
Invalid escape	\e (valid: \n \r \t \b \f)
Server-side JS	LiveWire (Netscape)
JS runs via	Script engine — TRUE
eval('x+y') with x=12,y=8	20
forEach sum [1,2,3]	6

The 20 Comparisons You Must Not Confuse

Pair	Quick Rule
id vs class	id = unique (#); class = shared (.)
div vs span	div = block (line break); span = inline (no break)
== vs ===	== value only; === value + type
let vs var vs const	let/const = block scope; var = function scope; const = no reassign
null vs undefined	null = intentional empty (typeof object); undefined = no value assigned
display:none vs visibility:hidden	none removes space; hidden keeps space
margin vs padding	margin = outside border; padding = inside border
GET vs POST	GET = data in URL; POST = data in body
absolute vs fixed	absolute = relative to ancestor; fixed = relative to viewport
relative vs absolute	relative = offset but keeps space; absolute = removed from flow

Pair	Quick Rule
em vs rem	em = parent font-size; rem = root font-size
strong vs b	strong = semantic; b = presentational
em vs i	em = semantic emphasis; i = presentational italic
article vs section	article = standalone; section = needs context
innerHTML vs innerText	innerHTML = parses HTML; innerText = plain text only
querySelector vs getElementById	querySelector = CSS selector; getElementById = id only
<svg> vs <canvas>	svg = vector; canvas = scripted raster
hover vs active	hover = mouse over; active = currently clicking
break vs continue	break = exit loop; continue = skip iteration
push() vs pop()	push = add to end; pop = remove from end

Memory Tricks

- LoVeHA: Link Visited Hover Active — correct CSS link pseudo-class order.
- MBPC (outside→inside): Margin Border Padding Content.
- HTML = nouns (what), CSS = adjectives (how it looks), JS = verbs (what it does).
- === is 'triple equals' — check THREE things: value, type, and nothing else.
- display: NONE = nothing (removed). visibility: HIDDEN = hidden ghost (space stays).
- GET = Gets data in URL (public). POST = Posts data in body (private).
- #id means 'One and only one'. .class means 'Many can share'.
- is STRONG importance. is just Bold. is EMphasis. <i> is just Italic.

★ EXAM TIP

All the best for your TCS IPA exam. Focus on: (1) output-tracing JS questions, (2) === vs ==, (3) display:none vs visibility:hidden, (4) which tags are HTML5, (5) class vs id selectors. These categories account for the majority of the UI section MCQs.